

Course / Programme:	BSc (Hons) Computing / BSc (Hons) Computer Science / BSc (Hons) Artificial Intelligence
Module Name:	Web Design and Programming
Module Code:	COM5409
Assignment Number & Title:	Assignment 1 – Full-stack web solution
Weighting:	100%
Issue Date:	10 March 2026
Submission Deadline:	Artefact and Decision Log via Moodle 23 April 2026 by 23:59 Demo in person 24 April 2026
Learning Outcomes:	LO1: Assess and outline issues and concerns surrounding web technologies and make appropriate recommendations for their mitigation. LO2: Analyse and evaluate an organisation's problems and requirements to design web-based solution with consideration of user experience (UX) and user interface (UI) design. LO3: Develop, implement and test a data-driven web-based solution using appropriate and contemporary web-development techniques, languages, tools, and frameworks and reflect on own performance

Assessment Task

This assignment requires you to design, develop, and reflect upon a full-stack web solution of your own choosing. You will apply the knowledge and skills gained throughout this module to build a data-driven web application using the module's technology stack.

You are free to choose the domain and scope of your application (e.g. a task manager, a recipe book, a student grade tracker, a blog platform, an inventory system, etc.). The more features and techniques you implement from the module content, the higher the grade you can achieve.

Technology Stack

Layer	Technology
Backend / API	Node.js with Express
Database	SQL or no-SQL with Prisma ORM
Frontend	React with JavaScript/TypeScript
Styling	CSS library (i.e.: Tailwind CSS)
Testing	Jest / React Testing Library / Playwright
Version Control	Git (with regular, meaningful commits)

Deliverables

Source Code

A zip file(.zip) of your complete project source code, including:

- All backend and frontend source files
- Prisma schema and migration files
- Any test files you have written
- A README.md file with setup instructions (how to install dependencies, configure the database, and run the application locally)

Important: Do not include node_modules or .env files. Your .env.example file should document the required environment variables.

Decision Log (max. 1000 words)

A short written document in which you justify the key technical and design decisions you made while building your application. This is not a formal essay, but rather a concise record of your reasoning. You should cover the following where relevant to your solution:

- **Technology and design choices:** Why did you structure your application the way you did? For example, why did you choose a particular database technology, a specific component structure, or a certain approach to state management? If you considered alternatives, briefly explain why you went with your chosen approach.
- **Web technology concerns:** What issues or risks did you identify with the technologies you used (e.g. security vulnerabilities, accessibility, performance)? What steps did you take to mitigate them?
- **UI/UX rationale:** How did you approach the design of your user interface? What informed your layout, navigation, and styling decisions? You may include simple wireframes or screenshots to support your reasoning.
- **Reflection:** What would you do differently next time?

You do not need to cover every decision—focus on the ones that mattered most. The quality of your reasoning

is more important than the length of your writing.

Additionally, include **a declaration of GAI tools** used appended to the decision log.

Keep in mind, references, figures, tables and the GAI declaration are **not** included in the word count.

Program Specification / Specific Assessment Criteria

You should implement the features and techniques described below in order. To be eligible for a mark within any classification, you must have at least attempted all the features for all the previous classifications.

Pass (40%-49%)

TypeScript does not need to be used for a Pass grade, however it is advised. You still have to use JavaScript.

Backend/API

- At least one working API endpoint for each CRUD operation (Create, Read, Update, Delete)
- API routes are connected to a database (SQL or no-SQL, excluding SQLite) via Prisma ORM
- Prisma schema defines at least one data model with appropriate field types

Frontend

- At least one React component that renders data retrieved from the API (e.g.: a table)
- A functional form that sends user input to the backend (e.g. creating/deleting/updating a record)

Styling & UI

- Basic use of a CSS library (i.e.: Tailwind) for layout and styling
- The application is usable and navigable (i.e. a user can complete core actions without confusion)

Code Quality

- Code runs without critical errors
- Project has a README.md with basic setup instructions
- Code is commented appropriately
- Git has been used with at least several commits showing progression

Lower Second Classification (50-59%)

You need to implement all of the above first before attempting the next features. Also, from this grade boundary onwards, you have to use **TypeScript**.

Backend/API

- Multiple related data models in Prisma (e.g. a one-to-many or many-to-many relationship)
- Meaningful error handling on API routes (e.g. returning appropriate HTTP status codes, handling not-found or invalid input scenarios)
- Input validation on at least some endpoints (e.g. checking required fields, data types)

Frontend

- Multiple React components with clear separation of responsibilities
- Effective use of React Hooks (useState, useEffect) for state management and data fetching

- User feedback on actions (e.g. loading indicators, success/error messages)

Styling & UI

- Consistent, considered visual design using Tailwind utility classes
- Evidence of basic UI/UX principles (e.g. clear navigation, logical layout, readable typography)

Architecture

- Clear project structure with separation of concerns (e.g. routes, controllers/handlers, and database logic are not all in one file)
- Use of middleware for at least one cross-cutting concern (e.g. logging, CORS, body parsing)

Upper Second Classification (60%-69%)

You need to implement all of the above first before attempting the next features.

Backend/API

- RESTful API design with consistent and meaningful route naming conventions
- API documentation using Swagger/OpenAPI or an equivalent approach
- Use of environment variables for configuration (e.g. database URL, port)

Frontend

- Well-structured component hierarchy (e.g. presentational vs. container components, reusable UI elements)
- Proper TypeScript usage throughout (typed interfaces/types for props, API responses, and state)
- Client-side form validation with meaningful error messages for users

Testing

- Unit tests for at least some backend logic (e.g. utility functions, validation) using Jest
- At least one component test using React Testing Library
- Evidence that tests were used to verify application behaviour (not just written after the fact)

Code Quality

- Regular, meaningful Git commit history demonstrating iterative development
- Clean, readable code with consistent naming conventions and sensible file organization

First Classification (70%-85%)

Security

- User authentication implemented (e.g. registration and login using JWT or sessions)
- Passwords are securely hashed (e.g. using bcrypt)
- Protected routes: certain API endpoints and/or frontend pages are restricted to authenticated users
- Input sanitisation to guard against common vulnerabilities (e.g. XSS, SQL injection)

Testing

- A comprehensive testing approach covering multiple layers of the testing pyramid

- Integration tests that verify backend endpoints work correctly with the database
- Component tests that verify meaningful user interactions (not just rendering)
- Evidence of a test-driven or test-informed development approach

Frontend

- Polished and professional user interface that demonstrates strong UI/UX design skills
- Effective handling of application state across multiple views or user workflows
- Accessibility considerations (e.g. semantic HTML, keyboard navigation, sufficient colour contrast)

Architecture

- Well-considered application architecture (e.g. layered/N-tier structure with clear responsibilities)
- Reusable middleware and/or utility modules that reduce code duplication
- Appropriate use of error-handling middleware for consistent API error responses

High First Class (85%-100%)

Security

- Demonstrates awareness and mitigation of multiple OWASP Top 10 risks beyond basic input validation
- Role-based access control or equivalent authorisation strategy
- Evidence of security-conscious design throughout the application (e.g. rate limiting, CORS configuration, secure headers)

Testing

- End-to-end tests using Playwright that cover key user workflows
- Near-comprehensive test coverage across backend and frontend
- Tests are well-organised, clearly named, and demonstrate excellent testing practices

Innovation

- Implementation goes beyond taught content, showing initiative and independent learning
- Exceptional code quality: clean, well-documented, and following industry best practices
- Creative and original solution with a high level of polish and attention to detail

Grading

A percentage mark will be provided based on General Assessment Guidelines for Written Assessments. Grading is as follows:

- A: 70 - 100%
- B: 60 - 69%
- C: 50 - 59%
- D: 40 - 49%

Marks below 40% will be classed as fail.

Secondary Research Requirements

Secondary research support is expected should be correctly cited using Harvard Referencing for both in-text citations and Reference Structure. For further details please see <https://www.bolton.ac.uk/leaponline/My->

Submission

Submission of both the decision log and the source code should be made to Turnitin via Moodle before the report deadline using the file naming convention below:

- COM5409_DecisionLog_IDnumber
- COM5409_Source_IDnumber

Declaration

At the end of the assessment, you should also include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment.

Secondary Research Requirements

Normally the expectation is that there would be a minimum of 10 sources, but since this is a practical assignment, it is understood that a smaller number may be used.

Use of Generative Artificial Intelligence (GAI) Applications in this Assessment

All submitted work must be your own. You must declare any use of Generative AI (GAI) tools in your submission. GAI may be used as a learning aid (e.g. to understand concepts, debug errors, understand the assignment brief, or explore approaches), but the design decisions, implementation, and written analysis must demonstrably be your own work. Submissions that are substantially AI-generated will be treated as Academic Misconduct. You are expected to be able to explain and justify any part of your code and Decision Log if asked.

AI Status	Application	Notes
Category B	GAI can be used to assist you with the assessment. <i>– Specify tasks</i>	Grammar and/or spell checkers may be used to correct individual words and sentences. GAI can also be used for specific tasks, e.g.: Debugging. <u>Any GAI generated content which is presented as your own original work and is not acknowledged will be assessed for academic misconduct.</u>

Guidelines for the Preparation and Submission of Written Assessments

Written assessments should be word-processed in Arial or Calibri Light font size 12. There should be double-spacing and each page should be numbered.

There should be a title page identifying the programme name, module title, assessment title, your student number, your marking tutor and the date of submission.

You should include a word-count (excluding reference list/bibliography, figures, tables and appendices).

Where a word limit is specified, the following penalty systems applies:

Up to 10% over the specified word length = no penalty

10 – 20% over the specified indicative word length = 5 marks subtracted (but if the assessment would normally gain a pass mark, then the final mark to be no lower than the pass mark for the assessment).

More than 20% over the indicative word length = if the assessment would normally gain a pass mark or more, then the final mark will be capped at the pass mark for the assessment.

At the end of the assessment you should include a declaration of any software tools including Generative AI (GAI) applications that you used in developing and completing the assessment. The assessment brief will specify if and how you can use GAI applications in the assessment.

All written work should be referenced using the standard University of Greater Manchester referencing style (**HARVARD**)– see: <https://libguides.bolton.ac.uk/resources/referencing/>

Unless otherwise notified by your Module Tutor, electronic copies of assignments should be saved as word documents and uploaded into Turnitin via the Moodle class area. If you experience problems in uploading your work, then you must send an electronic copy of your assessment to your Module Tutor via email BEFORE the due date/time.

Please note that when you submit your work to Moodle, it will automatically be checked for matches against other electronic information, as well as for hidden text characters and GAI generated text. You will be able to see similarity matches but not currently flags for hidden characters and AI-generated text. The outcomes of Turnitin reports may be used as evidence in an academic misconduct investigation (see Section 14).

Late work

Late work will be subject to the following penalties:

Up to 7 calendar days late = 10 marks subtracted but if the assignment would normally gain a pass mark, then the final mark to be no lower than the pass mark for the assignment.

More than 7 calendar days late = This will be counted as non-submission and no marks will be recorded.

Late submission of assessments on refer and those which are graded Pass/Fail only, is not permitted unless an extension is approved. See below.

Extensions

In the case of exceptional and unforeseen circumstances, an extension of up to 14 days after the assessment deadline may be requested using the standard University Extension Request Form. For approval there would need to be an explanation and evidence of relevant circumstances.

Longer extensions for individual assessments, projects and artefacts may be granted, at the discretion of the Programme Leader.

Requests for extensions which take a submission date past the end of the module (normally week 15) must be made using the Special Circumstances

Some students with registered disabilities will be eligible for revised submission deadlines. Revised submission deadlines for disability adjustments do not require the completion extension request paperwork. However, students should request these in writing in advance.

Please note that the failure of data storage systems is not considered to be a valid reason for an extension. It is therefore important that you keep multiple copies of your work on different storage devices before submitting it.

Academic Misconduct

Academic misconduct may be defined as any attempt by a student to gain an unfair advantage in any

assessment. This includes plagiarism, collusion, commissioning (contract cheating and unauthorised use of GAI) amongst other offences.

In order to avoid these types of academic misconduct, you should ensure that all your work is your own and that sources and software applications are attributed. You can also check originality through Turnitin.

Please note that penalties apply if academic misconduct is proven. See the following link for further details:
<https://greatermanchester.ac.uk/student-policy-zone/student-policies-2025-26>

GENERAL ASSESSMENT GUIDELINES – LEVEL HE5

	Relevance Learning outcomes must be met for an overall pass	Knowledge and Understanding	Analysis, Creativity and Problem-Solving	Self-awareness and Reflection	Research/ Referencing	Written English	Presentation and Structure
Class I (Exceptional Quality) 85% - 100%	Work is directly relevant and expertly addresses the requirements of the brief. Learning outcomes are met.	Demonstrates an exceptional breadth and depth of knowledge and understanding of theory and practice beyond the threshold expectation for the level. Demonstrates sophisticated conceptual understanding of specialised areas.	Presents a perceptive synthesis and critical evaluation of findings' drawing on a broad range of relevant sources in order to draw clear, systematic, justified conclusions. Provides a critical insight and interprets complex matters and ideas comprehensively. Demonstrates creative flair and a high level of originality. Demonstrates exceptional problem- solving skills and initiative.	Provides insightful reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility.	A wide range of contemporary and relevant reference sources selected and drawn upon. Sources cited accurately in both the body of text and in the Reference List/ Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. An exceptionally well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, effective inclusion of, and reference to, figures, tables and images.
Class I (Excellent Quality) 70% - 84%	Work is relevant and comprehensively addresses the requirements of the brief. Learning outcomes are met.	Demonstrates an excellent breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates in-depth understanding of key areas in different contexts.	Presents an excellent appraisal of findings through the critical analysis of information sources. Draws clear, systematic, justified and perceptive conclusions. Demonstrates creative flair, a high level of originality and initiative. Demonstrates a critical understanding of problem-solving approaches and applies excellent problem-solving skills.	Provides excellent reflection and self-awareness in relation to the outcomes of own work and personal responsibility.	A wide range of relevant reference sources selected and drawn upon. Sources cited accurately in both the body of text and in the Reference List/ Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. An excellently well written answer with competent, spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, effective inclusion of, and reference to, figures, tables and images.
Class II/i (Very Good Quality) 60% - 69%	Work is relevant and addresses most of the requirements of the brief. Learning outcomes are met.	Demonstrates a thorough breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates very good understanding of key areas in different contexts.	Presents a perceptive appraisal of findings through the critical analysis of information. Draws clear, justified and thoughtful conclusions. Demonstrates a very good level of creativity and originality. Demonstrates a critical understanding of problem-solving approaches and applies strong problem-solving skills.	Provides very good reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	A range of appropriate reference sources selected and drawn upon. Sources cited accurately in the main both the body of text and in the Reference List/Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. A very well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured in the main. Where relevant, effective inclusion of, and reference to, figures, tables and images.
	Relevance Learning outcomes must be met for an overall pass	Knowledge and Understanding	Analysis, Creativity and Problem-Solving	Self-awareness and Reflection	Research/ Referencing	Written English	Presentation and Structure

Class II/ii (Good Quality) 50% - 59%	Work addresses key requirements of the brief. Some irrelevant content. Learning outcomes are met.	Demonstrates a sound breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates sound understanding of key areas in different contexts.	Presents a logical appraisal of findings through the critical analysis of information. Draws clear and justified conclusions. Demonstrates creativity and originality. Demonstrates a critical understanding of problem-solving approaches and applies effective problem-solving skills.	Provides valid reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	Relevant reference sources selected and drawn upon. Most sources accurately cited both the body of text and in the Reference List/Bibliography.	Writing style is mostly appropriate to the requirements of the assessment Grammar, spelling and punctuation are generally competent and minor lapses do not pose difficulty for the reader. Paragraphs are structured and include some linking and signposting. Sentences are complete. A range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Logically structured in the most part. Inclusion of figures, tables and images but not all relevant or referred to.
Class III (Satisfactory Quality) 40% - 49%	Work addresses the requirements of the brief, although superficially in places. Some irrelevant content. Learning outcomes are met.	Demonstrates a sufficient breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates understanding of a range of key areas in different contexts.	Presents valid appraisal of findings in parts following a critical appraisal of information. Interprets a number of matters and ideas but with some descriptive passages evident which lack clear purpose. Demonstrates creativity and originality in places. Demonstrates a critical understanding of problem-solving approaches and applies sufficient problem- solving skills.	Provides some reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	Some relevant reference sources selected and drawn upon. Some weaknesses in referencing technique.	Writing style is occasionally not appropriate for the assessment. Grammar, spelling and punctuation are generally competent, but may pose minor difficulties for the reader. Some paragraphs may lack structure, and there is limited linking and signposting. Some appropriate vocabulary is used	The presentational style and layout are largely correct for the type of assignment. Adequately structured. Inclusion of some figures, tables and images but not all clear, relevant and/or referred to.
Borderline Fail 35% - 39%	Work addresses some of the requirements of the brief. Irrelevant and superficial content. One or more learning outcomes have not been met.	Demonstrates a lack of knowledge and understanding of theory and practice for this level. Demonstrates insufficient understanding of a range of areas in different contexts.	Presents a descriptive overview of findings following some analysis of information. Some passages lack clear purpose. Demonstrates scant creativity or originality. Demonstrates a lack of critical understanding of problem-solving approaches and insufficient problem-solving skills.	Provides limited reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.	Sources selected are limited and/or lack relevance. Poor referencing technique employed.	Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear in places. Paragraphs are poorly structured.	For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.
Fail <34%	Work does not address the requirements of the brief. Irrelevant and superficial content. One or more learning outcomes have not been met.	Demonstrates inadequate knowledge and understanding of theory and practice for this level. Demonstrates insufficient understanding of relevant areas.	Presents a descriptive overview of findings with little consideration of the quality of information drawn upon. Demonstrates little or no creativity. Demonstrates a lack of understanding of problem-solving approaches and limited problem- solving skills.	Provides inadequate reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.	An absence of academic and relevant sources selected and drawn upon. Poor referencing technique employed.	Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear. Unstructured paragraphs.	For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.